Blessed G/Michael Catholic Secondary School

Semester 1 Physics Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which of the following is a pure vector quantity?

A. Speed **B.** Mass **C.** Displacement **D.** Kinetic energy

____ **2.** Two vectors of magnitudes 3 N and 4 N act at a right angle (90 degrees) to each other. What is their resultant magnitude?

A. 1 N **B.** 5 N **C.** 7 N **D.** 12 N

____ **3.** A car accelerates uniformly from rest to 20 m/s in 5 seconds. What is its acceleration?

A. 2 m/s^2 **B.** 4 m/s^2 **C.** 5 m/s^2 **D.** 10 m/s^2

____ **4.** How far does a body travel during 4 seconds of free fall from rest? (Take g = 10 m/s^2)

A. 40 m **B.** 60 m **C.** 80 m **D.** 100 m

____ **5.** In projectile motion, what happens to the horizontal velocity component (neglecting air resistance)?

A. It increases continuously
B. It decreases continuously
C. It remains constant throughout the flight
D. It becomes zero at maximum height

____ **6.** What launch angle produces the maximum horizontal range for a projectile launched on level ground?

A. 30 degrees **B.** 45 degrees **C.** 60 degrees **D.** 90 degrees

____ **7.** Newton’s First Law of Motion is also known as the Law of:

A. Acceleration **B.** Conservation of Momentum
C. Inertia **D.** Gravitation

____ **8.** A net force of 50 N acts on an object of mass 10 kg. What is the resulting acceleration?

A. 0.2 m/s^2 **B.** 5 m/s^2 **C.** 50 m/s^2 **D.** 500 m/s^2

____ **9.** Action and reaction force pairs mentioned in Newton’s Third Law:

A. Act on the same object and cancel each other
B. Act on two different interacting bodies
C. Are unequal in magnitude when there is acceleration
D. Act in the same direction

____ **10.** The coefficient of static friction (mu_s) compared to kinetic friction (mu_k) for the same surfaces is generally:

A. Greater **B.** Smaller **C.** Equal **D.** Zero

____ **11.** A 1500 kg truck travelling at 20 m/s brakes to a complete stop in 4 seconds. What magnitude of braking force was applied?

A. 3,000 N **B.** 5,000 N **C.** 7,500 N **D.** 15,000 N

____ **12.** What is the work done when a force of 40 N moves a box 5 meters in the exact direction of the force?

A. 8 J **B.** 45 J **C.** 200 J **D.** 800 J